Workshop #2

Aeroservoelasticity of Fixed and Rotary Wing Aircraft

Francisco Javier Martín López
11072201

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Abstract

This report presents the aeroelastic analysis of the Puma blade, studying the influence of the aerodynamic model, the modal base of structural modes used and the influence of some parameters in the flutter stability.

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1 Pitch-Flap Flutter using Quasi-steady aerodynamics

Objective: Analyze the stability of the Puma blade considering only the first torsion and bending modes, employing a Quasi-steady aerodynamics model.

For this workshop, a code based on Ritz global functions has been developed, utilizing the same parameter interpolation for the Puma blade as in the provided reference code.After identifying the first bending and torsion orthogonal modes (approx. 4.6 Hz and 24.8 Hz respectively, corresponding to 29 rad/s and 156 rad/s), the aerodynamic forces are computed projecting the system onto this two-mode basis.

For the Quasi-Steady approximation, the Theodorsen function $C(k)$ is set to unity ($C(k) = 1$), assuming infinitesimal reduced frequencies.The resulting damping ratios (ζ) obtained for the modes are 0.64 and 0.245 respectively. As expected, both modes are stable, with the bending mode exhibiting significantly higher aerodynamic damping compared to the torsion mode.

2 Pitch-Flap Flutter using Unsteady Reference Section Aerodynamics

Objective: Analyze the stability of the Puma blade considering only the first torsion and bending modes, employing an unsteady aerodynamics model.

In this section, the Theodorsen function $C(k)$ has been evaluated at a representative reference section located at 75% of the blade span ($0.75R$), contrasting with the constant unity value ($C(k) = 1$) used in the previous Quasi-steady analysis.

The damping ratios obtained in this case are **0.58 and 0.241** for the bending and torsion modes respectively. This represents a reduction of approximately **10% and 1.5%** compared to the Quasi-steady case. This loss of damping is consistent with the physics of unsteady aerodynamics, as the magnitude of the Theodorsen function drops below unity ($|C(k)| < 1$) for non-zero reduced frequencies, reducing the effective aerodynamic damping forces.

3 Stability Map

Objective: Vary the pitch link stiffness and the position of the CG and plot the stability map, identifying the boundary between stable and unstable configurations.

The pitch link stiffness, modeled in the code as an additional term in the torsional virtual work equation, is swept between 0 and its nominal value. Simultaneously, the CG position is varied between 24% and 40% of the chord (measured from the leading edge), assuming a constant spanwise offset.

The resulting stability map is presented below:

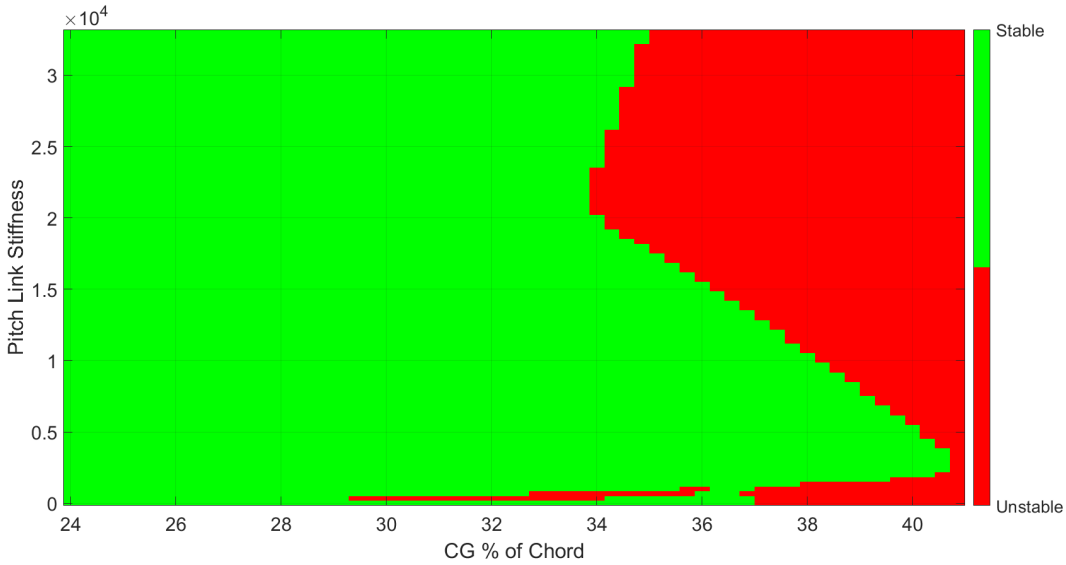


Figure 1: Stability Map: Pitch Link Stiffness vs CG position

Regarding the lower branch of the unstable margin, some irregularities are observed. Taking a look at the eigenvalues, it is seen that the instability in this region is a divergence, with near zero imaginary parts even in the marginally stable points there.

The map demonstrates that a proper design of the pitch link is necessary, as the stiffness peak around $2 \cdot 10^4$ N/m modifies the stability margin by 2-3% with respect to the nominal case. However, the location of the CG is the most critical parameter, as it is one of the main causes of mode coupling and, consequently, the primary driver of the flutter phenomenon.

4 Stability Map using Extended Modal Basis

Objective: Vary the pitch link stiffness and the position of the CG to plot the stability map, identifying the boundary between stable and unstable configurations, considering a higher number of structural modes.

In this section, the procedure employed is identical to the previous one, but the modal basis describing the structural motion has been expanded. Specifically, the first 10 modes have been included in the analysis.

The resulting stability map is presented below:

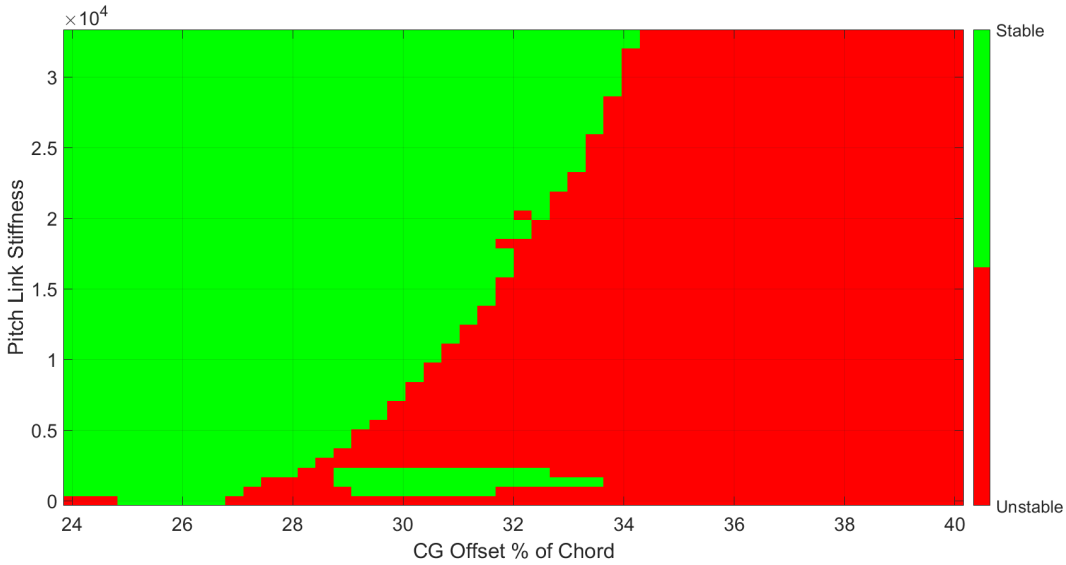


Figure 2: Stability Map: Pitch Link Stiffness vs CG position (10 Modes)

The main difference in this plot, compared to the simplified 2-mode analysis, is the expansion of the unstable region, which now contains a larger portion of the aft CG configurations. The inclusion of a higher number of modes captures additional coupling mechanisms between structural and aerodynamic forces, making the system more susceptible to flutter and thus expanding the unstable region.

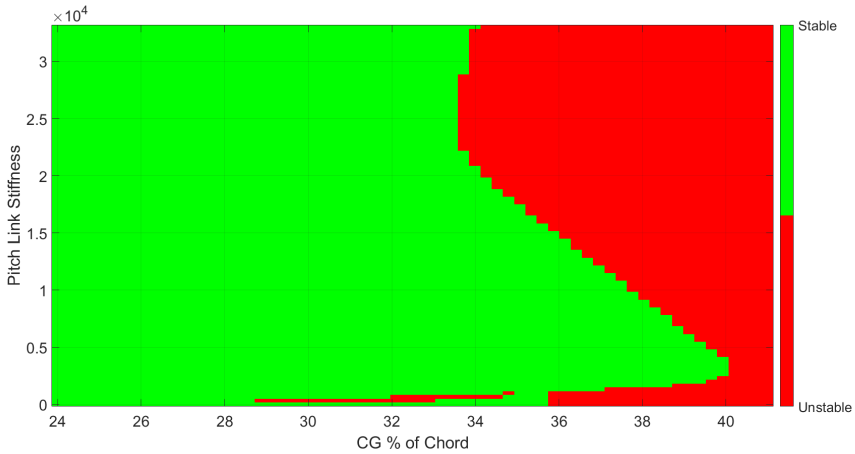
While the increased number of eigenvalues adds complexity to mode identification in certain regions, a significant observation is that the fifth mode exhibits very low damping (on the order of 0.01) even within the theoretically stable margin.

5 Stability Map with Exact k Unsteady Aerodynamics

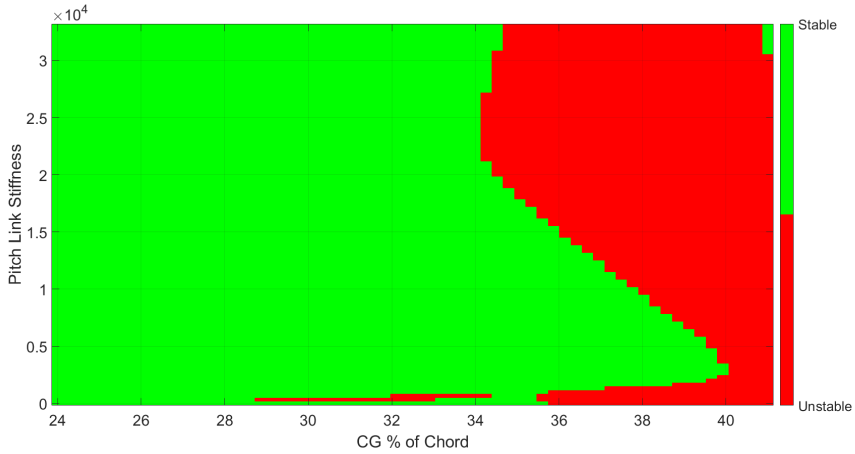
Objective: Computation of the unsteady aerodynamic forces using the local reduced frequency of each section.

To compute the aerodynamic forces with higher accuracy, the reduced frequency k and the corresponding Theodorsen function $C(k)$ are evaluated locally at each spanwise integration point, accounting for the variation of velocity along the blade span ($U = \Omega r$).

The stability maps from Section 3 (Reference Section at 0.75R) and the current analysis are compared below:



(a) Stability Map: Reference Section (from Section 3)



(b) Stability Map: Exact Local Reduced Frequency

Figure 3: Comparison of Stability Maps: Reference Section vs. Exact Local Unsteady Aerodynamics

The overall shape of the stability map remains unchanged between both methods. However, a slight shift of the unstable boundary to the right is observed in the upper zone of the 'exact' case map. This implies that the simplified approach using a characteristic section at 0.75R was conservative, as it predicted a slightly smaller stability margin than the more accurate spanwise integration.